

EC400: Solutions to Exam

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1. The optimization problem is

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}} \quad & \sum_{t=0}^{\infty} \beta^t \log (C_t - \theta C_{t-1}) \\ \text{s.t.} \quad & B_{t+1} = B_t - C_t, \end{aligned}$$

where B_0 and C_{-1} are given. Denote Lagrange multiplier with λ_t and write

$$\mathcal{L} = \sum_{t=0}^{\infty} \left[\beta^t \log (C_t - \theta C_{t-1}) + \lambda_t (B_t - C_t - B_{t+1}) \right].$$

The first-order conditions wrt C_t and B_{t+1} are

$$\begin{aligned} \beta^t \frac{1}{C_t - \theta C_{t-1}} - \beta^{t+1} \frac{\theta}{C_{t+1} - \theta C_t} &= \lambda_t, \\ \lambda_t &= \lambda_{t+1}. \end{aligned}$$

2. Combine together the previous equations to obtain

$$\frac{1}{C_t - \theta C_{t-1}} = \frac{\beta(1 + \theta)}{C_{t+1} - \theta C_t} - \frac{\beta^2 \theta}{C_{t+2} - \theta C_{t+1}}.$$

Given that λ_t is a shadow price of B_t , it follows that in a finite-period model, the terminal condition is $\lambda_T B_{T+1} = 0$. The infinite-period analog is

$$\lim_{t \rightarrow \infty} \beta^t \left(\frac{1}{C_t - \theta C_{t-1}} - \frac{\beta \theta}{C_{t+1} - \theta C_t} \right) B_{t+1} = 0.$$

Notice that since λ_t is constant and is generically different from zero, the TVC is

equivalen to $\lim_{t \rightarrow \infty} B_{t+1} = 0$.

3. At the beginning of period t , both B_t and $H_t = C_{t-1}$ are predetermined and the household can only choose C_t . Thus, the former two variables are states and the latter one is a control.

The steady-state version of the resource constraint implies that $\bar{C} = 0$. While the utility is equal minus infinity at this point, it is indeed the only feasible solution if $\bar{B} = 0$. Finally, the steady state habits are $\bar{H} = \bar{C} = 0$.

If the Blanchard-Kahn principle were satisfied, then the corresponding linearized dynamic system should have one eigenvalue greater than one and two eigenvalues less than one (in absolute terms). This implies that the steady state is a saddle.

4. The Bellman equation is given by

$$V(B, H) = \max_{0 \leq C \leq B} \left\{ \log(C - \theta H) + \beta V(B - C, C) \right\}.$$

5. Substitute the guess into the Bellman equation

$$V(B, H) = \max_C \left\{ \log(C - \theta H) + \beta \gamma \log(B - (1 + \phi)C) + \beta \alpha \right\}.$$

and take the first-order condition

$$\frac{1}{C - \theta H} = \frac{\beta \gamma (1 + \phi)}{B - (1 + \phi)C}.$$

Solve this equation for

$$C = \frac{1}{1 + \beta \gamma} \left(\frac{B}{1 + \phi} + \beta \gamma \theta H \right).$$

Substitute this expression back into the Bellman equation

$$\begin{aligned} V(B, H) &= \log \left(\frac{B - (1 + \phi)\theta H}{(1 + \beta \gamma)(1 + \phi)} \right) + \beta \gamma \log \left(\frac{\beta \gamma (B - \theta(1 + \phi)H)}{1 + \beta \gamma} \right) + \beta \alpha \\ &= (1 + \beta \gamma) \log(B - (1 + \phi)\theta H) + \text{const.} \end{aligned}$$

It follows that

$$\begin{aligned}\gamma = 1 + \beta\gamma &\Rightarrow \gamma = \frac{1}{1 - \beta}, \\ \phi = (1 + \phi)\theta &\Rightarrow \phi = \frac{\theta}{1 - \theta}\end{aligned}$$

and the policy function is $C(B, H) = (1 - \beta)(1 - \theta)B + \beta\theta H$.

6. The budget constraint for arbitrary period's length is $B_{t+\Delta} - B_t = -C_t\Delta$. Dividing both sides of the equation by Δ and taking the limit $\Delta \rightarrow 0$, we get $\dot{B}_t = -C_t$.

Notice that the law of motion for habits can be written as

$$H_{t+1} - H_t = C_t - C_{t-1}$$

and extends to the period of arbitrary length Δ :

$$H_{t+\Delta} - H_t = C_t\Delta - C_{t-\Delta}\Delta,$$

where we used the fact that C_t is a flow and H_t is a stock. Rewrite the equation using again $H_t = C_{t-\Delta}$ as

$$\frac{H_{t+\Delta} - H_t}{\Delta} = C_t - H_t$$

and take the limit $\Delta \rightarrow 0$ to get $\dot{H}_t = C_t - H_t$.

7. Denote with λ_t and μ_t two co-states associated respectively with the laws of motion for B_t and H_t . The Hamiltonian is then given by

$$\mathcal{H}_t = e^{-\rho t} \log(C_t - \theta H_t) - \lambda_t C_t + \mu_t (C_t - H_t)$$

Use the maximum principle:

$$\begin{aligned}
0 &= \frac{\partial \mathcal{H}_t}{\partial C_t} = \frac{e^{-\rho t}}{C_t - \theta H_t} - \lambda_t + \mu_t, \\
-\dot{\lambda}_t &= \frac{\partial \mathcal{H}_t}{\partial B_t} = 0, \\
-\dot{\mu}_t &= \frac{\partial \mathcal{H}_t}{\partial H_t} = -\frac{\theta e^{-\rho t}}{C_t - \theta H_t} - \mu_t, \\
0 &= \lim_{t \rightarrow \infty} \lambda_t B_t, \\
0 &= \lim_{t \rightarrow \infty} \mu_t H_t.
\end{aligned}$$

8. The fact that $\lambda_t = \lambda$ follows immediately from the second optimality condition above. Conjecture that $\dot{\mu}_t = 0$. From the third condition it follows that $\mu_t = -\frac{\theta e^{-\rho t}}{C_t - \theta H_t}$ and we need to check that the latter ratio is indeed constant. To this end, differentiate the first optimality condition with respect to time

$$-\frac{\rho e^{-\rho t}}{C_t - \theta H_t} - \frac{e^{-\rho t}(\dot{C}_t - \theta \dot{H}_t)}{(C_t - \theta H_t)^2} - \dot{\lambda}_t + \dot{\mu}_t = 0.$$

Use the fact that both co-states are constant in time and simplify to get the Euler equation:

$$\frac{\dot{C}_t - \theta \dot{H}_t}{C_t - \theta H_t} = -\rho.$$

Solve this integral equation

$$\log(C_t - \theta H_t) = \text{const} - \rho t \quad \Rightarrow \quad C_t - \theta H_t = \text{const} \cdot e^{-\rho t}.$$

Substituting back into the expression for μ_t , we confirm the initial guess that μ_t is constant.¹

9. Go back to the Bellman equation and rewrite it for a period of arbitrary length as

$$V(B_t, H_t) = \max_{C_t} \left\{ \log(C_t - \theta H_t) \Delta + e^{-\rho \Delta} V(B_{t+\Delta}, H_{t+\Delta}) \right\}.$$

¹Showing that there are no other solutions requires additional effort and the transversality conditions.

Put all terms on one side and use the approximation for the discount factor:

$$\max_{C_t} \left\{ \log (C_t - \theta H_t) \Delta + (1 - \rho \Delta) V(B_{t+\Delta}, H_{t+\Delta}) - V(B_t, H_t) \right\} = 0.$$

Add and subtract $V(B_t, H_{t+\Delta})$ and combine flows and stocks:

$$\begin{aligned} \max_{C_t} \left\{ \log (C_t - \theta H_t) \Delta - \rho V(B_{t+\Delta}, H_{t+\Delta}) \Delta \right. \\ \left. + V(B_{t+\Delta}, H_{t+\Delta}) - V(B_t, H_{t+\Delta}) + V(B_t, H_{t+\Delta}) - V(B_t, H_t) \right\} = 0. \end{aligned}$$

Divide all terms by Δ :

$$\begin{aligned} \max_{C_t} \left\{ \log (C_t - \theta H_t) - \rho V(B_{t+\Delta}, H_{t+\Delta}) + \frac{V(B_{t+\Delta}, H_{t+\Delta}) - V(B_t, H_{t+\Delta})}{B_{t+\Delta} - B_t} \frac{B_{t+\Delta} - B_t}{\Delta} \right. \\ \left. + \frac{V(B_t, H_{t+\Delta}) - V(B_t, H_t)}{H_{t+\Delta} - H_t} \frac{H_{t+\Delta} - H_t}{\Delta} \right\} = 0. \end{aligned}$$

Take the limit $\Delta \rightarrow 0$ and simplify to get the HJB equation:

$$\rho V(B, H) = \max_C \left\{ \log (C - \theta H) + V_B(B, H) \dot{B} + V_H(B, H) \dot{H} \right\}$$

or equivalently,

$$\rho V(B, H) = \max_C \left\{ \log (C - \theta H) - V_B(B, H) C + V_H(B, H) (C - H) \right\}.$$